Problem Set 21

Volumes of Revolution II: Other Axes

October 27, 2026

Problem 9

(a)

$$S = \int_0^3 3x - x^2\,dx = \frac{3}{2}x^2 - \frac{x^3}{3}\Big|_0^3 = \boxed{\frac{9}{2}}$$

(b)

$$V = \pi \int_0^3 (4x - x^2 + 2)^2 - (x+2)^2\,dx$$

$$= \pi \int_0^3 x^4 + 16x^2 + 4 - 4x^2 + 16x - 8x^3 - x^2 - 4x - 4\,dx$$

$$\pi(\frac{x^5}{5} - 2x^4 + \frac{11}{3}x^3 + 6x^2)\Big|_0^3$$

$$= \boxed{\frac{198\pi}{5}}$$

(c)

$$V = \pi \int_0^3 \left(2 + \sqrt{4-y}\right)^2 - (4-y)^2 dy$$

$$+\pi \int_3^4 \left(2 + \sqrt{4-y}\right)^2 - \left(2 - \sqrt{4-y}\right)^2 dx$$

(d)

$$W(k) = \pi \int_0^3 (f(x)+k)^2 - (g(x)+k)^2 dx$$

$$W'(k) = 2\pi \int_0^3 f(x) - g(x)\,dx$$

$$= \boxed{9\pi}$$

(e)

$$V = 2\pi(k-y)A$$

$$\frac{198\pi}{5} = 2\pi(y+2)\frac{9}{2}$$

$$22 = 5y + 10 \implies y = \frac{12}{5}$$

$$k = \frac{12}{5} + \frac{22}{5} = \boxed{\frac{34}{5}}$$

Problem 10

(a)

$$T = \int_0^2 \frac{4}{4+x^2}\,dx = 4 \cdot \frac{1}{2} \arctan 1 = \boxed{\frac{\pi}{2}}$$

(b)

$$\begin{aligned} T &= \pi \int_0^2 \left(\frac{4}{4+x^2}\right)^2 dx \\ &= \pi \int_0^2 \frac{16}{(4+x^2)^2}\,dx \\ &= 16\pi \int_0^{\frac{\pi}{4}} \frac{2\sec^2\theta\,d\theta}{16\sec^4\theta} \\ &= \pi \int_0^{\frac{\pi}{4}} 2\cos^2\theta\,d\theta \\ &= \pi \int_0^{\frac{\pi}{4}} \cos 2\theta + 1\,d\theta \\ &= \pi(\frac{\sin 2\theta}{2} + \theta)\Big|_0^{\frac{\pi}{4}} \\ &= \boxed{\frac{\pi^2}{4} + \frac{\pi}{2}} \end{aligned}$$

(c)

$$\begin{aligned} y = \frac{4}{4+x^2} \implies x &= 2\sqrt{\frac{1}{y} - 1} \\ V &= \pi \int_0^{\frac{1}{2}} 4\,dy + \pi \int_{\frac{1}{2}}^1 \frac{4}{y} - 4\,dy \\ &= 2\pi + \pi(4\ln y)\Big|_{\frac{1}{2}}^1 - 2\pi \\ &= \boxed{4\pi \ln 2} \end{aligned}$$

(d)

$$\begin{aligned} V &= \pi \int_0^2 1 - (1 - \frac{4}{4+x^2})^2\,dx \\ &= \pi \int_0^2 2 \cdot \frac{4}{4+x^2} - \left(\frac{4}{4+x^2}\right)^2 dx \\ &= 2\pi \cdot \frac{\pi}{2} - \frac{\pi^2}{4} - \frac{\pi}{2} \\ &= \boxed{\frac{3\pi^2}{4} - \frac{\pi}{2}} \end{aligned}$$

(e)

$$T = \int_0^b \frac{4}{4+x^2}\,dx = 4 \cdot \frac{1}{2}\arctan\frac{b}{2} = \frac{\pi}{4}$$
$$b = \boxed{2\tan\left(\frac{\pi}{8}\right)}$$

(f) The volume of the piece with $0 \le x \le b$ is larger, as more distance is traveled by the centroid.

$$V = \pi\int_0^{\frac{\pi}{8}} \cos 2\theta + 1\,d\theta$$
$$= \pi(\frac{\sin 2\theta}{2} + \theta)\Big|_0^{\frac{\pi}{8}}$$
$$= \boxed{\frac{\pi^2}{8} + \frac{\sqrt{2}\pi}{4}}$$

Problem 11

(a)

$$x - y = 0$$
$$y = x^2 \implies x - x^2 = 0$$
$$d = \boxed{\frac{x-x^2}{\sqrt{2}}}$$
$$X - x^2 = -(X - x)$$
$$X = \frac{x^2+x}{2}$$
$$u = \boxed{\frac{x^2+x}{\sqrt{2}}}$$

(b)

$$V = \int_0^{\sqrt{2}} \pi d^2\,du$$
$$= \pi\int_0^1 \left(\frac{x-x^2}{\sqrt{2}}\right)^2 \cdot \frac{2x+1}{\sqrt{2}}\,dx$$
$$= \frac{\pi}{2\sqrt{2}}\int_0^1 2x^5 - 3x^4 + x^2\,dx$$
$$= \frac{\pi}{2\sqrt{2}}(\frac{2x^6}{6} - \frac{3x^5}{5} + \frac{x^3}{3})\Big|_0^1$$
$$= \boxed{\frac{\pi}{30\sqrt{2}}}$$

(c)

$$A = \int_0^1 x - x^2\,dx = \frac{x^2}{2} - \frac{x^3}{3}\Big|_0^1 = \frac{1}{6}$$

$$\delta = \frac{\frac{1}{2} - \frac{2}{5}}{\sqrt{2}} = \frac{1}{10\sqrt{2}}$$

$$\frac{\pi}{30\sqrt{2}} = 2\pi \cdot \frac{1}{10\sqrt{2}} \cdot \frac{1}{6}$$

$$\boxed{V = 2\pi\delta A}$$

(d)

$$A = 1$$

$$\delta = \frac{\frac{3}{2} + \frac{1}{2}}{\sqrt{2}} = \sqrt{2}$$

$$V = 2\pi \cdot \sqrt{2} \cdot 1 = \boxed{2\pi\sqrt{2}}$$